

**Large-Scale Quantitative Characterization of Human Induced Pluripotent Stem Cell Colony
Morphology Using Image Processing and Unsupervised Machine Learning**

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Abstract

Human induced pluripotent stem cell (hiPSC) colony morphology is routinely evaluated during cell culture, yet conventional assessment relies heavily on visual inspection and may be subjective and difficult to standardize. Quantitative image analysis provides an opportunity to characterize colony morphology objectively and at large scale. In this study, I developed a computational framework for large-scale quantitative analysis of hiPSC colony morphology using publicly available segmented microscopy data from the Allen Institute for Cell Science. A total of 2,890 segmentation images were processed, yielding 69,714 individual colony objects from 2,474 images represented in the final colony-level dataset. Nine morphological features describing colony size and geometry were extracted automatically from segmentation masks. Standardized features were analyzed using principal component analysis (PCA), K-Means clustering, and statistical testing. The first two principal components explained 91.76% of the total morphological variance, with the first component primarily representing colony size and the second capturing shape-related characteristics including eccentricity, solidity, and extent. Unsupervised clustering identified two morphological groups comprising 68,204 and 1,510 colonies, respectively. The groups showed pronounced differences in size-related features, with particularly large effect sizes for area, perimeter, axis lengths, convex area, and equivalent diameter. An image-level analysis additionally identified a moderate negative correlation between colony number and mean colony area ($r = -0.369$). These findings demonstrate that large-scale quantitative morphology analysis can reveal substantial and interpretable structure within hiPSC colony populations without requiring predefined biological labels. The resulting framework provides a reproducible foundation for future studies linking quantitative colony morphology with experimentally validated biological states.

Keywords: hiPSC, induced pluripotent stem cells, colony morphology, image analysis, machine learning, PCA, K-Means clustering, quantitative morphology

1. Introduction

Human induced pluripotent stem cells (hiPSCs) have become an important experimental system for regenerative medicine, disease modeling, drug discovery, and developmental biology because they can be expanded and differentiated into a wide range of specialized cell types. However, maintaining consistent hiPSC cultures requires regular assessment of cell and colony characteristics. Colony morphology is commonly used as a practical indicator during culture because changes in colony size, shape, boundary organization, and cellular arrangement can provide information about the state of a culture. Despite its widespread use, morphological assessment is often based on visual inspection, making it potentially subjective and difficult to standardize across experiments and laboratories.

Previous studies have demonstrated that quantitative image analysis can provide objective measurements of pluripotent stem cell morphology. Kato et al. developed a quantitative framework for analyzing human pluripotent stem cell colony morphology using multiple morphological parameters and hierarchical clustering, demonstrating that computational measurements could identify distinct morphological characteristics within cell colonies [1]. Other studies have investigated machine learning approaches for automated assessment of pluripotent stem cell colonies. Fan et al. developed a label-free imaging and machine learning framework for detecting and predicting induced pluripotent stem cell colony formation during somatic cell reprogramming [2]. Additional studies have applied conventional machine learning and deep learning approaches to classify pluripotent stem cell colonies according to predefined morphological or quality categories [3–5].

These studies demonstrate the potential of image-derived morphology as a scalable and non-invasive source of information about pluripotent stem cell cultures. However, many existing approaches focus on supervised classification, in which colonies are assigned predefined categories, or on predicting specific outcomes such as colony formation or culture quality. Although these approaches are valuable when

reliable biological labels are available, they can limit the ability to explore morphological variation without imposing predefined categories.

Unsupervised analysis provides an alternative approach in which morphological patterns can be identified directly from image-derived measurements without requiring prior biological labels. This approach is particularly useful for exploratory characterization, where the objective is to determine which morphological characteristics account for the greatest variation within a population and whether recurring morphological phenotypes can be identified.

The increasing availability of large-scale microscopy datasets provides an opportunity to perform such analyses at a substantially larger scale than would be practical through manual assessment. The Allen Institute for Cell Science has developed large-scale imaging resources for human induced pluripotent stem cells, including colony-level overview imaging and associated segmentation data [6]. These resources provide an opportunity to investigate colony-level morphology computationally using publicly accessible data.

In this study, I developed a computational framework for large-scale quantitative characterization of hiPSC colony morphology using segmented microscopy images from the Allen Institute for Cell Science. I extracted individual colony objects from segmentation masks and calculated quantitative morphological features describing colony size and geometry. I then applied feature standardization, principal component analysis (PCA), K-Means clustering, and statistical analysis to characterize the structure of morphological variation within the dataset. The analysis included 69,714 individual colony objects and examined both colony-level and image-level morphological characteristics.

The primary objective of this study was not to assign biological identities to the identified clusters, but rather to characterize the structure of morphological variation within a large hiPSC colony population. By combining interpretable image-derived morphological measurements with unsupervised machine learning, I aimed to establish a scalable analytical framework for identifying recurring morphological

phenotypes that could provide a foundation for future studies linking quantitative colony morphology with experimentally validated biological states.

2.1 Dataset

This study utilized publicly available human induced pluripotent stem cell (hiPSC) microscopy data obtained from the Allen Institute for Cell Science through the Allen Cell Imaging Collections. The selected dataset, the hiPSC 12X Overview Image Dataset, provides microscopy images and associated segmentation data for quantitative analysis of human cell morphology.

The dataset contains segmented microscopy images of hiPSC colonies, in which individual colony regions are represented by unique label values within segmentation masks. Background pixels are represented by a value of zero, while each detected colony is assigned a unique integer label. The availability of segmentation masks enabled automated identification and extraction of individual colony objects without requiring manual annotation.

For this study, 2,890 segmented TIFF images were processed using an automated image-analysis pipeline. A total of 2,886 images were successfully processed, while four images failed during processing due to image-reading or segmentation-related issues, corresponding to a successful processing rate of approximately 99.9%. The processing pipeline resulted in the extraction of 69,714 individual colony objects.

The resulting colony-level dataset was subsequently used for morphological feature extraction, dimensionality reduction, unsupervised clustering, and statistical evaluation of morphological differences between identified colony groups.

2.2 Image Processing and Feature Extraction

The segmentation images provided by the Allen Institute were processed using an automated image analysis pipeline developed in Python. Each TIFF segmentation mask contained labeled regions corresponding to individual hiPSC colonies, where background pixels were assigned a value of zero and each detected colony was represented by a unique integer label.

For each segmentation image, connected colony regions were identified and analyzed using the `regionprops_table` function from the `scikit-image` library. This approach enabled automated extraction of quantitative morphological measurements from every detected colony while maintaining consistency across the complete dataset.

A total of nine morphological features were extracted for each colony: area, perimeter, eccentricity, solidity, extent, major axis length, minor axis length, convex area, and equivalent diameter. These features were selected to capture both colony size characteristics and geometric shape properties.

Area, perimeter, major axis length, minor axis length, convex area, and equivalent diameter were used to describe colony size and spatial dimensions. Eccentricity quantified colony elongation, while solidity and extent provided information regarding colony compactness, boundary irregularity, and occupancy within the bounding region.

The extracted measurements from all processed images were combined into a single colony-level dataset containing the image identifier, colony label, and corresponding morphological features. This resulted in a final dataset of 69,714 individual hiPSC colonies available for downstream statistical analysis and machine learning-based characterization.

2.3 Morphological Feature Analysis

Prior to statistical analysis and machine learning, extracted morphological features were standardized to ensure that measurements with larger numerical ranges did not dominate downstream analyses. Feature scaling was performed using standard normalization, transforming each feature to have a mean of zero and a standard deviation of one.

The standardized feature matrix was used for exploratory analysis of morphological variation among hiPSC colonies. Principal component analysis (PCA) was applied to reduce feature dimensionality while preserving the maximum amount of variance within the dataset. PCA enabled identification of the dominant morphological characteristics contributing to variation between colonies and provided a lower-dimensional representation for visualization.

The contribution of individual morphological features to the principal components was evaluated through analysis of PCA loading values. This allowed interpretation of whether observed morphological variation was primarily associated with colony size, geometric properties, or shape-related characteristics.

2.4 Dimensionality Reduction and Clustering

Unsupervised machine learning was used to identify recurring morphological patterns among hiPSC colonies without requiring predefined biological labels. Principal component analysis (PCA) was first performed on the standardized morphological feature matrix to reduce dimensionality and visualize the distribution of colony morphologies in a lower-dimensional space.

The first two principal components were used for visualization, capturing 91.76% of the total variance within the dataset. Analysis of PCA loading values was performed to determine the contribution of individual morphological features to the observed variation.

K-Means clustering was subsequently applied to the standardized morphological feature space to identify groups of colonies exhibiting similar morphological characteristics. Different numbers of clusters were evaluated using the elbow method and silhouette analysis to assess clustering quality and separation between groups.

The final clustering solution of two clusters was selected based on the combined evaluation of clustering metrics and the interpretability of the resulting morphological separation. Solutions with additional clusters produced less meaningful subdivisions, including extremely small groups likely representing outliers rather than distinct morphological populations.

The identified clusters were subsequently characterized by comparing their morphological feature distributions and calculating statistical differences between groups.

2.5 Statistical Analysis

Statistical analyses were performed to evaluate differences in morphological characteristics between the identified colony clusters. Since morphological feature distributions were non-normal and contained substantial variation across thousands of observations, non-parametric statistical testing was used.

The Mann–Whitney U test was applied independently to each extracted morphological feature to determine whether significant differences existed between the two identified clusters. Statistical significance was assessed using a threshold of $p < 0.05$.

In addition to statistical significance testing, effect sizes were calculated using Cohen's d to quantify the magnitude of morphological differences between clusters. This provided a measure of practical significance beyond p -values, particularly given the large sample size of the dataset.

Image-level relationships between colony abundance and morphology were also evaluated. Pearson correlation analysis was performed to examine the association between the number of detected colonies per image and the mean colony area within each image.

All analyses were performed using Python-based scientific computing libraries, including NumPy, pandas, scikit-learn, scikit-image, SciPy, and visualization libraries.

3.1 Large-Scale Dataset Characterization

A large-scale morphological dataset was generated from segmented hiPSC colony images obtained from the Allen Institute for Cell Science. In total, 2,890 segmentation TIFF images were processed using the automated image-analysis pipeline, resulting in the extraction of 69,714 individual colony objects. The final colony-level dataset contained morphological information from 2,474 unique microscopy images.

Before clustering, the resulting dataset consisted of 11 variables, including the image identifier, colony label, and nine quantitative morphological features describing colony size and geometry. A twelfth variable, representing the assigned cluster, was added during the clustering analysis. The extracted colonies showed substantial morphological variation, with colony area ranging from 360 to 1,318,639 pixels and equivalent diameter ranging from 21.41 to 1,295.74 pixels.

Across all detected colonies, the mean colony area was 14,680.72 pixels with a standard deviation of 61,331.55 pixels, indicating considerable heterogeneity in colony size. The mean number of colonies detected per image was 28.18 ± 13.26 , with substantial variation between images reflecting differences in colony abundance.

These results demonstrate the scale and morphological diversity of the dataset and establish a foundation for subsequent dimensionality reduction, clustering, and statistical characterization of hiPSC colony morphology.

3.2 Principal Component Analysis Reveals Dominant Morphological Variation

Principal component analysis (PCA) was performed to investigate the major sources of variation within the extracted hiPSC colony morphology features. The first two principal components accounted for 91.8% of the total variance within the dataset, with PC1 explaining 77.6% and PC2 explaining 14.2% of the observed morphological variation.

Analysis of PCA loadings revealed that PC1 was primarily associated with colony size-related characteristics. Features including area (loading = 0.363), perimeter (0.388), major axis length (0.388), minor axis length (0.391), convex area (0.376), and equivalent diameter (0.393) showed strong positive contributions to the first principal component. This indicates that the dominant source of variation among hiPSC colonies was related to overall colony dimensions. In contrast, PC2 was mainly influenced by shape-related morphological properties. Eccentricity showed a strong negative contribution to PC2 (-0.560), while extent (0.569) and solidity (0.461) contributed positively.

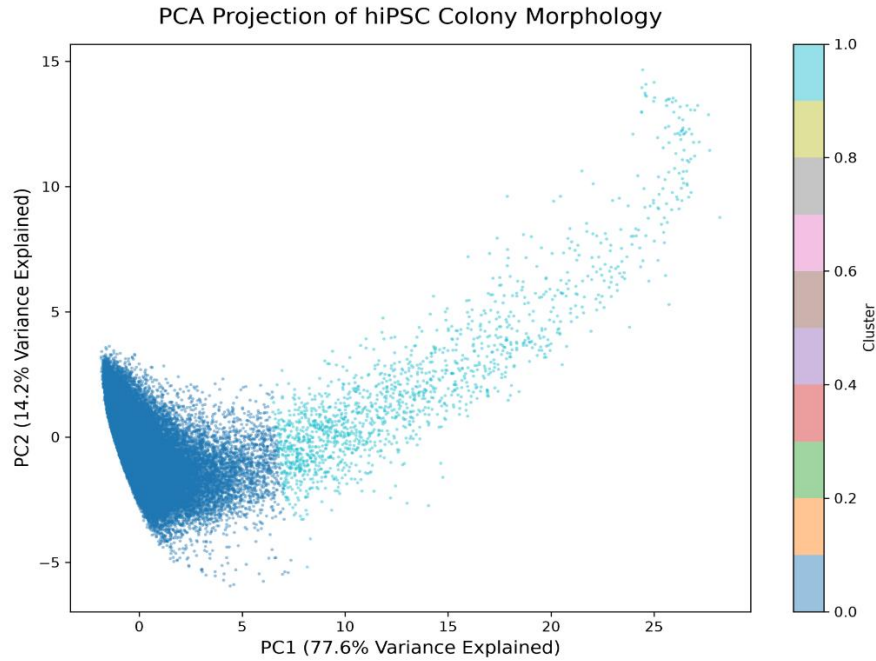


Figure 1. Principal component analysis of hiPSC colony morphology. The first two principal components explain 77.6% and 14.2% of the total variance, respectively.

These results suggest that the second principal component captured differences related to colony geometry, compactness, and boundary characteristics rather than overall colony size.

The PCA projection demonstrated that colony morphology could be represented effectively within a two-dimensional feature space, providing a basis for subsequent unsupervised clustering analysis

3.3 Unsupervised Clustering Identifies Two Distinct Morphological Phenotypes

K-Means clustering was applied to the standardized morphological feature dataset to identify groups of colonies with similar morphological characteristics. Different values of K were evaluated using the elbow method and silhouette analysis, together with consideration of the interpretability of the resulting groups. Based on these evaluations, a two-cluster solution was selected.

The resulting clusters were highly imbalanced. Cluster 0 contained 68,204 colonies (97.83%), while Cluster 1 contained 1,510 colonies (2.17%). Despite the substantial difference in cluster size, the two groups exhibited pronounced differences across several morphological measurements.

Cluster 0 represented the dominant morphological group and was characterized by relatively small colonies. The mean colony area was 7,504.79 pixels, with a mean perimeter of 374.86 pixels. The mean major and minor axis lengths were 119.09 pixels and 64.60 pixels, respectively, while the mean equivalent diameter was 77.98 pixels.

Cluster 1 represented a substantially smaller group characterized by much larger colony dimensions. The mean colony area increased to 338,804.77 pixels, while the mean perimeter increased to 7,043.99 pixels. Mean major and minor axis lengths were 1,061.16 pixels and 632.68 pixels, respectively, and the mean equivalent diameter was 623.34 pixels.

Differences were also observed in shape-related measurements. Cluster 0 exhibited higher mean solidity (0.866) and extent (0.579), whereas Cluster 1 exhibited lower mean solidity (0.578) and extent (0.395). Mean eccentricity was similar between the groups, with values of 0.759 for Cluster 0 and 0.763 for Cluster 1.

The clustering results therefore indicate that the primary distinction between the two morphological groups was associated with colony scale, accompanied by differences in compactness and geometric organization. The smaller Cluster 1 should be interpreted as a rare morphological phenotype rather than

as a biological cell state, since no independent biological labels were available to validate such an interpretation.

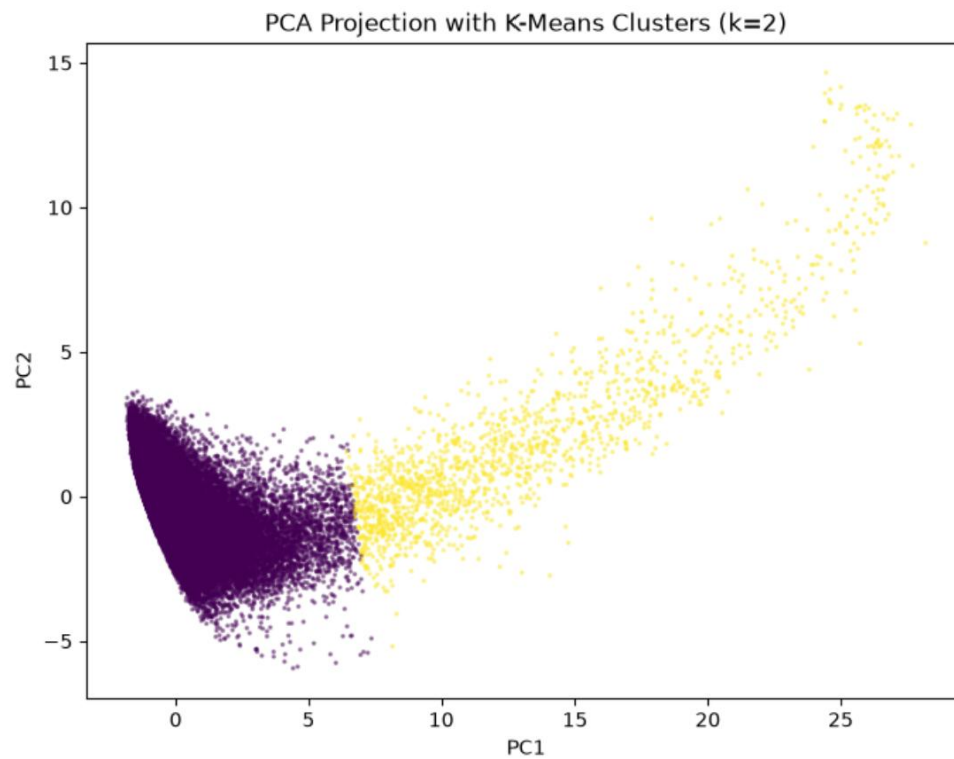


Figure 2. PCA projection of hiPSC colony morphology colored according to K-Means cluster assignment. Each point represents an individual colony, with colors indicating membership in Cluster 0 or Cluster 1. The projection illustrates the separation of the two morphological groups within the principal component space.

3.4 Statistical Validation of Cluster Differences

To evaluate whether the observed morphological differences between the two clusters were statistically supported, Mann–Whitney U tests were performed independently for each of the nine extracted morphological features. All nine features showed statistically significant differences between Cluster 0 and Cluster 1 ($p < 0.05$).

The strongest differences were observed for the size-related features. Colony area, perimeter, major axis length, minor axis length, convex area, and equivalent diameter all showed extremely large effect sizes, indicating that the differences between the two clusters were not only statistically significant but also substantial in magnitude.

Cohen's d values were -8.74 for area, -9.98 for perimeter, -8.09 for major axis length, -8.66 for minor axis length, -10.62 for convex area, and -8.29 for equivalent diameter. These values indicate exceptionally large differences between the morphological distributions of the two clusters.

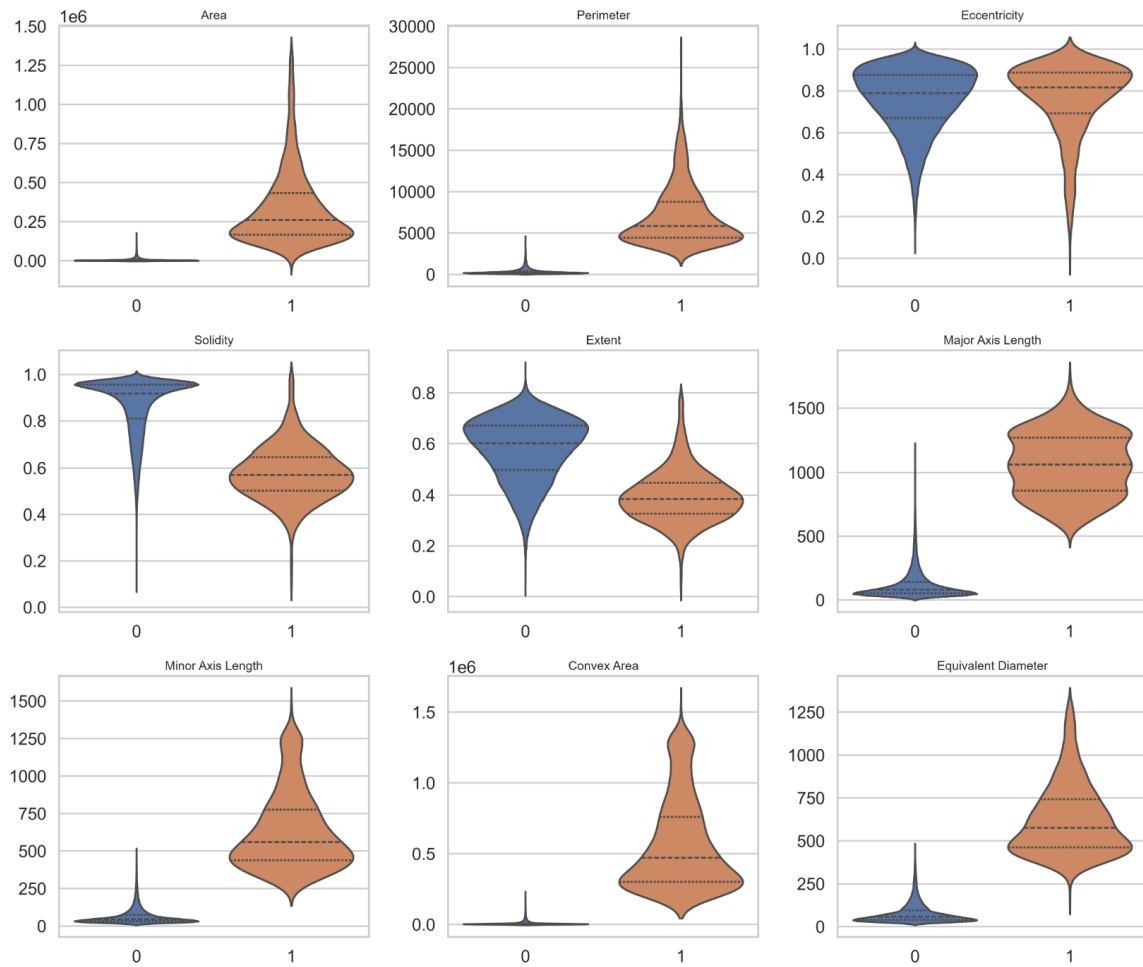
Shape-related features also showed statistically significant differences. Solidity produced a Cohen's d of 2.39, while extent produced an effect size of 1.52, indicating substantial differences in colony compactness and spatial occupancy between the two groups.

In contrast, eccentricity showed a much smaller effect size (Cohen's $d = -0.029$) despite being statistically significant ($p = 0.000019$). This indicates that the large sample size allowed a small difference in eccentricity to reach statistical significance, while the magnitude of the difference itself was negligible.

The similar eccentricity distributions observed in the cluster comparison therefore suggest that eccentricity was not a major characteristic distinguishing the two morphological groups.

Overall, the statistical analysis supports the interpretation from the PCA and clustering analyses that colony size was the dominant characteristic separating the two groups, while solidity and extent contributed additional differences related to colony shape and compactness

Morphological Feature Comparison Between Clusters

**Figure 3. Morphological feature distributions across K-Means clusters.**

Violin plots show the distributions of nine quantitative morphological features for Cluster 0 and Cluster 1, with internal quartile markers. Strong separation is observed for colony size-related features, including area, perimeter, major and minor axis length, convex area, and equivalent diameter. Solidity and extent also differ substantially between clusters, whereas eccentricity shows considerable overlap.

3.5 Image-Level Morphological Variation

In addition to colony-level analysis, I examined morphological characteristics at the image level to investigate variation in colony abundance and average colony size across microscopy images. For each image represented in the final dataset, the number of detected colonies, mean colony area, and within-image variability in colony area were calculated.

Across the 2,474 images represented in the final colony-level dataset, the mean number of detected colonies per image was 28.18 ± 13.26 , with values ranging from 1 to 75 colonies per image. This variation indicates substantial differences in colony abundance across the microscopy images.

To investigate the relationship between colony abundance and colony size, Pearson correlation analysis was performed between the number of detected colonies per image and the mean colony area. A moderate negative correlation was observed ($r = -0.3693$), indicating that images containing larger numbers of colonies tended to have smaller mean colony areas.

This relationship suggests that colony abundance and average colony size were not independent characteristics within the dataset. Images with higher colony counts generally contained smaller colonies, whereas images with fewer detected colonies tended to contain larger colonies. However, the moderate magnitude of the correlation indicates that colony abundance alone does not explain the variation in mean colony area.

Image-level analysis therefore provided an additional perspective on the morphological organization of the dataset beyond individual colony measurements. The observed inverse relationship between colony abundance and mean colony area identifies a potentially important population-level pattern that could be investigated further in future studies.

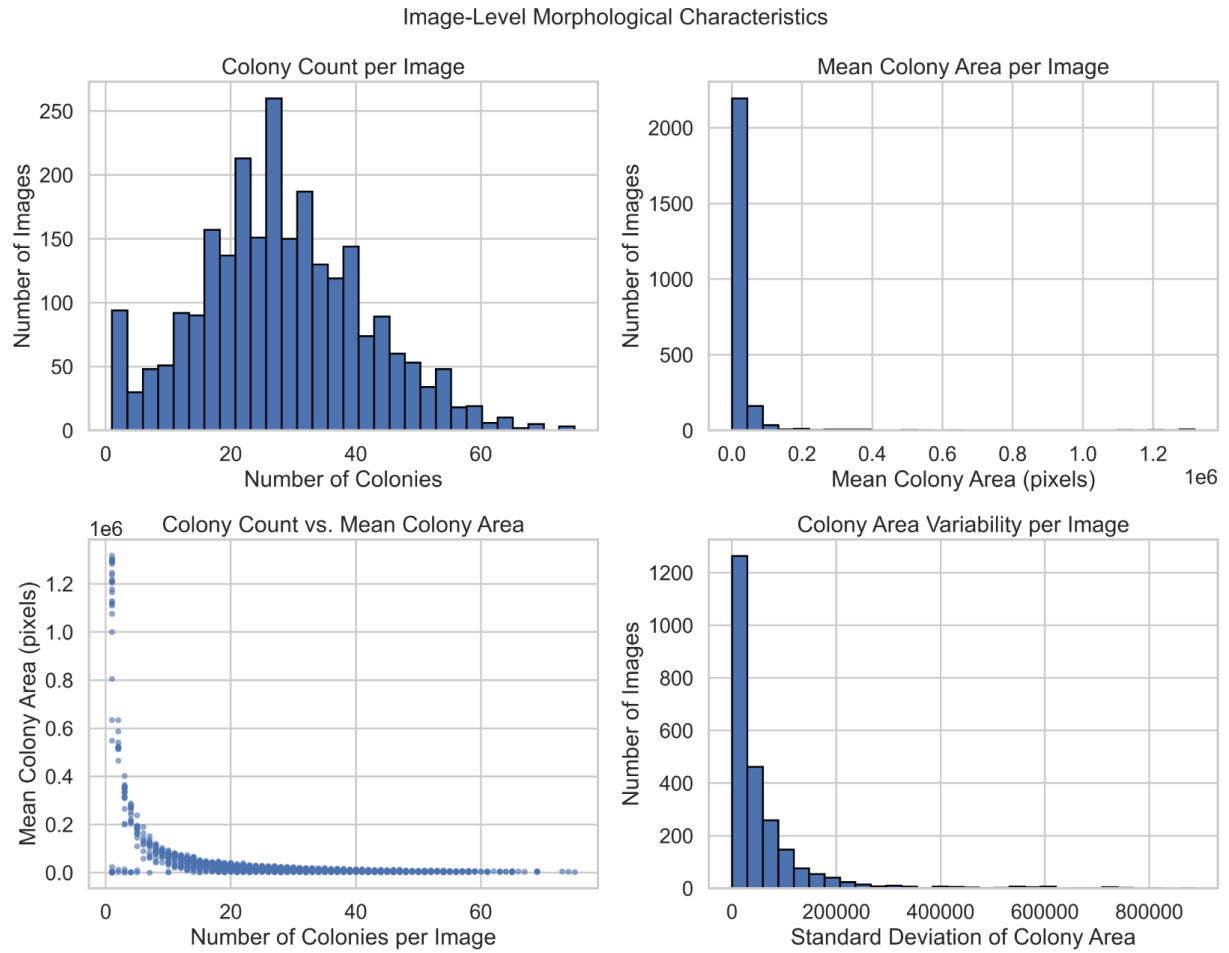


Figure 4. Image-level analysis of hiPSC colony morphology across microscopy images.

3.6 Representative Colony Morphologies

A representative segmented hiPSC colony is shown in Figure 5 to illustrate the morphological objects used for quantitative analysis. The colony was extracted directly from the segmentation mask by isolating an individual labeled colony and cropping the corresponding region with a fixed padding. In the resulting binary representation, the colony is shown as a white foreground object against a black background. The example segmented hiPSC colony exhibits an irregular, approximately rounded morphology with variations in its boundary, including local protrusions and indentations. These boundary characteristics are captured quantitatively by the morphological features extracted from the segmentation masks, including area, perimeter, eccentricity, solidity, extent, major and minor axis length, convex area, and equivalent diameter. This visualization demonstrates how individual hiPSC colonies were represented computationally prior to feature extraction and subsequent statistical and clustering analyses. Rather than relying solely on visual assessment, the image-processing pipeline converted colony morphology into quantitative measurements that could be compared across thousands of individual colonies.



Figure 5. Representative segmented hiPSC colony used for morphological feature extraction.

4.1 Quantitative Characterization of hiPSC Colony Morphology

The analysis of 69,714 individual hiPSC colonies demonstrated substantial variation in colony morphology across the dataset. This variation was captured through a combination of size-, shape-, and boundary-related measurements, including area, perimeter, eccentricity, solidity, extent, major and minor axis length, convex area, and equivalent diameter. The range of values observed across these features indicates that hiPSC colonies do not exhibit a single uniform morphological configuration, but instead occupy a diverse morphological space.

The use of quantitative measurements provides an objective means of characterizing this variation. While visual assessment can identify broad differences in colony appearance, converting colony morphology into numerical features allows these differences to be measured consistently across a large number of samples. This is particularly relevant for hiPSC cultures, where changes in colony size and shape may occur across different regions of a culture or between individual colonies.

The observed morphological heterogeneity also provides a basis for subsequent multivariate analysis. Because the extracted features describe different aspects of colony geometry, considering them jointly allows patterns of variation to be identified that may not be apparent from any single measurement. The PCA and clustering analyses presented in the subsequent sections therefore build upon this initial characterization by examining how individual colonies are distributed within the overall morphological feature space.

Importantly, the present analysis demonstrates morphological heterogeneity rather than establishing a direct relationship between a specific morphological phenotype and an underlying biological state. Establishing such relationships would require additional biological measurements or experimentally validated labels. Nevertheless, the ability to quantify morphology across tens of thousands of individual

colonies provides a foundation for investigating whether image-derived morphological patterns can serve as useful indicators of hiPSC culture characteristics in future studies.

4.2 Principal Component Analysis and Morphological Variation

Principal component analysis provided a reduced-dimensional representation of the morphological feature space and demonstrated that the variation among hiPSC colonies could be summarized through a smaller number of principal components. The distribution of colonies within the principal component space indicates that colony morphology is characterized by continuous variation rather than a single uniform phenotype.

The contribution of multiple morphological features to the principal components further suggests that differences between colonies arise from combinations of size and shape characteristics rather than from a single dominant measurement. Features related to colony dimensions and area contribute to the overall morphological structure, while shape descriptors provide additional information regarding differences in colony geometry.

The PCA representation also provided an important basis for the subsequent clustering analysis. By reducing the correlated morphological measurements into a lower-dimensional representation, PCA facilitated visualization of the overall structure of the dataset and allowed patterns of similarity and variation between individual colonies to be examined more clearly.

However, the PCA results should primarily be interpreted as a description of morphological variation within the dataset rather than evidence of distinct biological states. Principal components represent mathematical combinations of the measured features and do not independently correspond to specific biological processes. Further experimental validation would therefore be required to determine whether

the morphological patterns identified through PCA are associated with meaningful differences in hiPSC biological state or culture quality.

4.3 Morphological Clustering and Identification of Distinct Colony Phenotypes

The K-means analysis identified two groups of colonies with distinct positions within the standardized morphological feature space. The resulting clusters were highly unequal in size, with Cluster 0 containing 68,204 colonies and Cluster 1 containing 1,510 colonies. This distribution indicates that the dominant morphological pattern represented the majority of colonies, while a substantially smaller group exhibited a combination of morphological characteristics that distinguished it from the dominant population.

The identification of a smaller morphological group is potentially important because relatively uncommon colony morphologies may be difficult to identify consistently through visual inspection alone. Quantitative clustering provides a means of detecting such patterns across a large dataset and may therefore be useful for identifying colonies that warrant further investigation.

However, the unequal cluster sizes also require cautious interpretation. The smaller cluster should not automatically be interpreted as representing an abnormal or biologically distinct hiPSC population. K-means clustering identifies groups based on similarity in the selected morphological features, and the resulting separation may reflect differences in colony size, shape, or other geometric properties rather than a confirmed biological phenotype.

The clustering results therefore provide evidence for structured morphological heterogeneity within the dataset. Rather than establishing that two biologically distinct hiPSC states exist, the analysis demonstrates that the measured morphological characteristics can be used to identify a dominant

morphological pattern and a smaller, quantitatively distinct subgroup. Experimental validation using independent biological measurements would be necessary to determine whether this morphological separation corresponds to differences in cell state, pluripotency, differentiation, or culture quality.

4.4 Relationship Between Colony Abundance and Colony Size

The image-level analysis identified a moderate negative correlation between colony abundance and mean colony area across the 2,474 microscopy images, with a Pearson correlation coefficient of $r = -0.3693$. This indicates that images containing greater numbers of detected colonies generally tended to have smaller average colony areas, whereas images containing fewer colonies tended to contain larger colonies.

This relationship suggests that colony abundance and colony size may be related characteristics within the analyzed dataset. One possible explanation is that differences in the spatial distribution or degree of colony expansion across images may contribute to this inverse relationship. However, the correlation alone does not establish the underlying cause of the observed pattern. Factors such as culture conditions, spatial variation within the imaging field, colony growth dynamics, or segmentation-related effects could potentially contribute to the relationship.

The moderate magnitude of the correlation is also important. Colony count does not fully explain variation in mean colony area, indicating that additional factors contribute to differences in colony morphology between images. Therefore, colony abundance should be considered as an additional image-level characteristic rather than as a direct predictor of colony size.

Taken together with the colony-level analysis, these findings demonstrate that morphological variation can be examined at multiple levels of organization. Individual colony measurements characterize local

morphological differences, while image-level measurements reveal broader patterns in colony abundance and average size across microscopy fields. This multi-level perspective may be useful in future studies investigating how colony organization changes under different culture or experimental conditions.

4.5 Implications for Non-Invasive hiPSC Characterization

The findings of this study demonstrate the potential of image-based quantitative morphology as a non-invasive approach for characterizing variation within hiPSC cultures. The analysis transformed segmented microscopy images into quantitative measurements for tens of thousands of individual colonies, allowing morphological differences to be examined systematically rather than relying exclusively on qualitative visual assessment.

The combination of feature extraction, PCA, clustering, and image-level analysis provides several complementary perspectives on colony morphology. PCA demonstrated the overall structure of morphological variation, while K-means identified a smaller group of colonies with quantitatively distinct morphological characteristics. The image-level analysis further showed that colony abundance was associated with average colony size across microscopy images.

These findings suggest that computational morphology could potentially contribute to automated monitoring of hiPSC cultures. In a future application, image-derived measurements could be combined with experimentally validated indicators of culture quality to investigate whether specific morphological patterns can provide early indications of changes in cell state.

However, the present study does not establish morphology as a validated proxy for biological state. The results demonstrate that morphological information can be quantified and analyzed at scale, but a direct connection between the identified morphological patterns and biological characteristics remains to be

established. Such validation would be an important step toward developing morphology-based monitoring systems for practical hiPSC culture applications.

4.6 Limitations and Future Directions

Several limitations should be considered when interpreting the findings of this study. First, the analysis was based on image-derived morphological measurements obtained from segmentation masks. Consequently, the accuracy of the extracted features depends on the quality of the underlying segmentation. Errors in identifying colony boundaries could influence measurements such as area, perimeter, solidity, and shape-related features.

Second, the clustering analysis was unsupervised and was based exclusively on morphological features. Although the $K = 2$ solution provided a useful representation of the observed morphological structure, the resulting clusters cannot be assumed to correspond to biologically distinct populations without independent validation. This is particularly important given the substantial imbalance between the two clusters.

Third, the dataset did not provide direct biological labels that could be used to determine whether specific morphological patterns correspond to differences in pluripotency, differentiation status, viability, or other biological characteristics. Consequently, the present study identifies morphological patterns rather than assigning biological meaning to those patterns.

Future work should address these limitations by integrating image-derived morphology with experimentally measured biological variables. Potential validation strategies could include comparison with pluripotency markers, differentiation status, viability measurements, or other established indicators of hiPSC culture quality. Longitudinal imaging could also be used to determine whether morphological changes precede measurable changes in biological state.

Further methodological development could additionally incorporate more advanced image-analysis approaches, including texture-based features, spatial organization measurements, time-dependent morphological analysis, or deep-learning-based representations. Validation on independent datasets would also be important to determine whether the morphological patterns identified in the present dataset generalize across different imaging conditions, cell lines, and culture protocols.

4.7 Overall Interpretation

Overall, this study demonstrates that quantitative image analysis can capture substantial morphological heterogeneity among hiPSC colonies at both the individual-colony and image levels. Analysis of 69,714 colonies revealed variation across multiple geometric characteristics, while PCA provided a lower-dimensional representation of this morphological space and K-means identified a smaller subgroup with quantitatively distinct characteristics.

The image-level analysis further demonstrated that colony abundance and mean colony area were moderately inversely related across the analyzed microscopy images. Together, these findings indicate that hiPSC colony morphology contains structured quantitative variation that can be extracted from microscopy images using computational methods.

The primary contribution of the study is therefore not the identification of a specific biological phenotype, but the demonstration of a scalable framework for quantifying and exploring hiPSC colony morphology. By combining image segmentation, morphological feature extraction, dimensionality reduction, clustering, and image-level analysis, the approach provides a foundation for future studies investigating whether morphological features can serve as reliable, non-invasive indicators of hiPSC biological state and culture quality.

5. Conclusion

This study presented a quantitative image-analysis framework for characterizing the morphology of human induced pluripotent stem cell (hiPSC) colonies. Using segmented microscopy images, morphological features were extracted from 69,714 individual colonies across 2,474 microscopy images, enabling colony morphology to be examined at both the individual-colony and image levels.

The analysis demonstrated substantial morphological heterogeneity across the dataset. Principal component analysis provided a reduced representation of this variation, while K-means clustering identified two morphologically distinct groups, including a dominant population and a substantially smaller subgroup. Image-level analysis additionally revealed a moderate negative relationship between colony abundance and mean colony area, indicating that these characteristics were related but not fully dependent on one another.

Importantly, the findings demonstrate the feasibility of converting hiPSC colony morphology into quantitative measurements that can be analyzed systematically at large scale. Although the identified morphological patterns cannot be directly interpreted as biological states without experimental validation, the results provide a foundation for investigating whether image-derived morphology can serve as a non-invasive indicator of hiPSC culture characteristics.

Future studies should integrate quantitative morphology with independent biological measurements and longitudinal imaging to determine whether specific morphological patterns can reliably predict changes in cell state, pluripotency, differentiation, or culture quality. Such validation could ultimately support the development of automated image-based approaches for monitoring and characterizing hiPSC cultures.